

PARLA CHIARO: A Dialect-Aware Misunderstanding Alert System for AI Pipelines on Italian Regional Speech

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1 Introduction

Italy is characterized by a linguistically unique landscape in Europe: alongside standard Italian, the country encompasses dozens of regional varieties and dialects, Neapolitan, Sicilian, Venetian, Lombard, Sardinian, and many others, spoken daily by millions of people [1]. This richness represents both a cultural heritage and an open technological challenge: current Automatic Speech Recognition (ASR) systems and Large Language Models (LLMs) are trained predominantly on standard language varieties, exhibiting significantly higher error rates when faced with dialectal speech [2, 3].

The problem is amplified by the typical processing chain of modern voice-based AI assistants: the user's speech is first converted into text by an ASR model, and the resulting transcript is then interpreted by an LLM. Errors introduced at the transcription stage, substitutions, insertions, or deletions caused by dialectal phonology, Italian, dialect code-switching, or regional vocabulary with no standard Italian equivalent, are silently propagated to the LLM, which interprets them with unwarranted confidence and without alerting the user to a potential misunderstanding [4].

This project is situated within the **PARLA CHIARO** initiative, a research project of the University of Naples Federico II selected under the LINGUA Open Call of Microsoft for the preservation of Italian dialects in the era of Artificial Intelligence¹.

Students are asked to design and implement *DialectGuard*: a modular alert system that inserts itself into the ASR $\rightarrow$ LLM pipeline and (i) analyses the ASR transcript to detect indicators of potential misunderstanding caused by dialectal input, (ii) estimates a risk score, and (iii) triggers a user-facing alert mechanism, e.g. generating clarifying questions, flagging low-confidence text spans, or requesting reformulation, before the LLM produces a potentially misleading response.

¹<https://news.microsoft.com/source/emea/features/il-progetto-parla-chiaro-del-luniversita-di-napoli-federico-ii-selezionato-nellambito-della-lingua-open-call-di-microsoft-per-la-tutela-dei-dialetti-italiani-nellera-dellintelligenza-art>

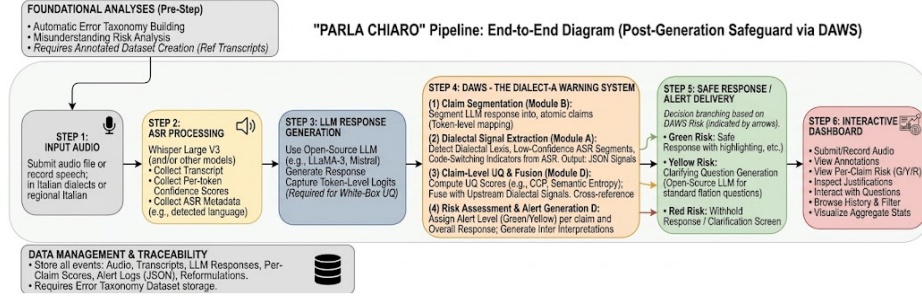


Figure 1: An example of pipeline for DialectGuard

2 Methodology

The project follows a modular pipeline aligned with a post-generation architecture:

audio $\rightarrow$ ASR $\rightarrow$ LLM $\rightarrow$ DAWS $\rightarrow$ [safe response / alert]

Each component contributes complementary signals to identify dialect-induced misunderstandings.

- **ASR Error Analysis and Dialectal Signal Extraction:**

Students analyse ASR outputs on dialectal speech (e.g. Neapolitan) to build an error taxonomy including phonetic substitutions, lexical distortions, code-switching, and prosodic errors. A subset of data is manually annotated to produce reference transcriptions and labeled errors. These annotations are used to train a detection module that identifies:

- dialectal lexical items,
- low-confidence ASR segments,
- code-switching patterns.

- **Claim Segmentation:**

The LLM response is decomposed into atomic claims, each corresponding to a verifiable statement. The segmentation must preserve token-level alignment, enabling downstream uncertainty estimation.

- **Claim-Level Uncertainty Quantification:**

For each claim, uncertainty is computed using one or more methods:

- white-box approaches (e.g. token likelihood, entropy, CCP);
- consistency-based methods (e.g. semantic entropy across multiple generations).

These methods assign a reliability score to each claim.

- **Dialect-Aware Risk Assessment:**
DAWS combines claim-level uncertainty with upstream ASR signals. Claims overlapping with dialectal or low-confidence ASR segments are assigned higher risk. The system outputs:
 - per-claim risk levels (green / yellow / red),
 - overall response risk score,
 - interpretable justifications.
- **Clarifying Question Generation:**
When the risk exceeds a threshold, the system generates clarification prompts using an open-source LLM. Questions must be concise, relevant, and formulated in standard Italian.
- **End-to-End Integration:**
All components are integrated into a modular pipeline interoperable with different ASR systems (e.g. Whisper, WhisperX) and LLMs (e.g. LLaMA-3, Mistral, Qwen).
- **Storage and Traceability:**
All system events are stored in a NoSQL database, including ASR outputs, detected signals, LLM responses, claim-level scores, and user interactions, enabling auditing and iterative improvement.
- **Interactive Dashboard:**
A user interface allows interaction with the system, including audio input, visualization of ASR signals, claim-level risk highlighting, clarification interaction, and aggregate analytics.

3 Objective

The main objective of this project is to design a scalable and interpretable Dialect-Aware Warning System (DAWS) that reduces the risk of misleading AI responses in the presence of dialectal input.

Unlike traditional approaches that attempt to detect errors before LLM inference, DAWS operates **post-generation**, analysing the actual claims produced by the LLM. The system must:

- Segment the LLM response into individual claims;
- Estimate uncertainty at the claim level using token-level or semantic methods;
- Integrate upstream ASR signals (e.g. low-confidence segments, dialect markers, code-switching);
- Compute a risk score for each claim and for the overall response;

- Trigger appropriate user-facing alerts, including highlighting unreliable claims, generating clarification questions, or withholding the response.

The system should focus on Southern Italian dialects (e.g. Neapolitan and Sicilian), while remaining generalizable to other varieties. Both quantitative evaluation (e.g. detection of unreliable claims) and qualitative assessment (e.g. usability and clarity of alerts) are required.

3.1 Practical Examples

- **Example 1: Voice-Based Health Assistant**

A user says: *“Tengo ’o dolore ’e cape, pozzo piglà ’a tachipirì?”*. The ASR produces a distorted transcript. The LLM generates a response that includes a claim referring to an incorrect condition.

DAWS segments the response into claims and identifies that one claim has high uncertainty and overlaps with low-confidence ASR segments. The system assigns a red risk level to that claim and triggers a clarification: *“I may have misunderstood some dialectal expressions. Did you mean: “I have a headache, can I take Tachipirina?”*

- **Example 2: Public-Service Chatbot**

A Sicilian query produces a heavily distorted transcript. The LLM generates multiple claims, most of which are assigned high uncertainty.

DAWS assigns an overall red risk level and withholds the response, prompting the user to rephrase the request.

3.2 Managed Data

The system processes and manages the following types of data:

- *Audio/Voice Recordings* in Italian dialects or strongly regionally accented Italian (sourced from public corpora or self-collected, see Section 5).
- *ASR Transcripts* with per-token confidence scores.
- *Dialectal Error Taxonomy Dataset*: an annotated dataset built by students during the analysis phase.
- *Risk Scores and Alert Logs*: structured JSON objects with metadata (detected dialect variety, error type, confidence thresholds, timestamp).
- *Interaction History*: complete user-system sessions including reformulations and final LLM responses, used for quantitative analysis.

For audio data collection, students may refer to the corpora listed in Section 5 and may also record or synthesize samples using open-source dialectal TTS tools.

4 Related Work and Starting Points

Students are encouraged to study and build upon the following works:

- **Ramponi (2024)** [1]: a comprehensive overview of Italy’s linguistic varieties, the NLP challenges they pose, and available corpora. Covers the state of the art for Neapolitan, Sicilian, Venetian, and other varieties, and advocates for a shift from machine-centric to speaker-centric NLP, the conceptual foundation of this project.
- **Survey on Spoken Italian Datasets** [2]: an analysis of 66 spoken Italian datasets, with dedicated coverage of regional and dialectal varieties (VIVALDI corpus, KIParla, etc.), essential for selecting audio resources for the project.
- **Fatapour Y. et al. (2025)** [3]: a systematic study of Whisper and WhisperX errors on non-native speakers in a clinical setting, showing that post-processing with GPT-4o substantially recovers transcription accuracy. The proposed “chained model” pipeline (ASR $\rightarrow$ LLM correction) is directly applicable to the dialectal context of this project.
- **Harris et al. (2024)** [4]: modeling of gender and dialect bias in ASR systems, with quantitative WER analysis across minority dialects. Provides benchmarking methodologies transferable to the Italian context.
- **GenSEC Challenge (2024)** [6]: a shared task on generative LLM-based correction of ASR errors, covering post-ASR transcription correction, speaker tagging, and emotion recognition. Directly reusable baselines and evaluation metrics for the correction module.
- **Aliannejadi et al. (2019)** [7] and **Zamani et al. (2020)** [8]: foundational works on clarifying question generation in conversational information-seeking systems. The Qulac and ClariQ datasets are starting points for the clarification generation module.
- **Kuhn et al. (2023)** [9]: a method for semantic uncertainty quantification in LLMs via linguistic invariances, useful for estimating the reliability of an LLM response downstream of a potentially distorted transcript.
- **LLM-Based ASR Technical Report (2025)** [10]: an overview of speech LLM architectures with focus on robustness to dialects, accents, and code-switching in low-resource settings, including retrieval-augmented ASR strategies.

5 Tools and Technologies

- **Speech Recognition:** Whisper Large V3 / WhisperX²; speech-LLM models (SLAM-ASR, Qwen-Audio) for advanced experiments.

²<https://github.com/openai/whisper>

- **Open-Source LLMs:** LLaMA-3, Mistral, Qwen, Gemma (via Hugging Face³); GGUF-quantized models via Ollama for local inference.
- **Dialectal Lexical Resources:** VIVALDI corpus⁴; Mozilla Common Voice Italian⁵; VoxPopuli IT⁶; KIParla corpus.
- **NLP and Classification:** Hugging Face Transformers; Camoscio / Fauno (Italian open-source LLMs); UmBERTo, AnITA for Italian NLP tasks.
- **Uncertainty Quantification:** calibration libraries (e.g. `uncertainty-baselines`); ASR logit entropy analysis.
- **Database Management:** MongoDB or ChromaDB for alert event storage and vector indexing.
- **Interactive Dashboard:** Streamlit, Dash, or Gradio.
- **Benchmark Datasets:** Mozilla Common Voice IT, FLEURS IT, dialectal audio datasets assembled during the project.

6 Project Evaluation

The project will be evaluated by a committee, which will assign a score from 0 to 5 to various aspects of the methodology and final dashboard. The evaluation of the methodology may vary based on the oral discussion and will be communicated only after the discussion.

6.1 Dashboard Evaluation

- **Usability:** the interface must be accessible even to users with low technological familiarity; alerts must be clear and not unnecessarily alarming.
- **Transparency:** risk spans must be visible and motivated; the user must understand *why* an alert was triggered.
- **Accuracy:** risk scores and error type labels must be correct and contextually relevant in the chosen application scenarios.
- **Efficiency:** the system must respond within acceptable latency; clarifying questions must be concise and actionable.
- **Customization:** users should be able to configure alert thresholds; the system should adapt to a user’s dialect variety over time.
- **Aggregate Analysis:** the dashboard must include visualizations of error history and distribution by dialect variety and error type, supporting ongoing quality monitoring.

³<https://huggingface.co/models>

⁴<https://www2.hu-berlin.de/vivaldi/>

⁵<https://commonvoice.mozilla.org/it>

⁶<https://github.com/facebookresearch/voxpathuli>

6.2 Methodology Evaluation

- **Error Taxonomy Quality:** completeness, linguistic grounding, and practical utility of the dialectal error classification built during the analysis phase.
- **Detection Module Performance:** quantitative evaluation on at least one benchmark (Precision, Recall, F1 for detecting dialect-distorted transcripts); false positive / false negative trade-off analysis.
- **Clarifying Question Quality:** assessment of the relevance and clarity of generated questions, via automatic metrics and/or human evaluation.
- **End-to-End Impact:** the extent to which the system reduces the rate of misleading LLM responses compared to a no-alert baseline.
- **Scalability:** the system’s ability to handle different dialect varieties and growing interaction volumes.
- **Reproducibility and Open Science:** use of open-source models; availability of code, configurations, and instructions to reproduce results.

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